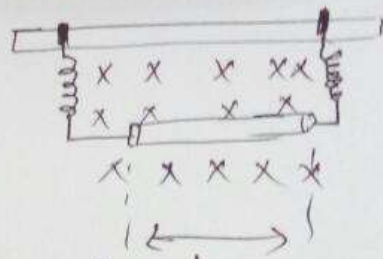


P-41: A 13g wire of length $L = 62.0\text{ cm}$ is suspended by a pair of flexible leads in a uniform magnetic field of magnitude 0.440 T (Fig. 28-41). What are the (a) magnitude and (b) direction (left or right) of the current required to remove the tension in the supporting leads?

$$m = 13\text{ g}$$

$$L = 62\text{ cm}$$

$$B = 0.440\text{ T}$$



(a) $i = ?$

(b) direction of current L

Req. to remove the tension in the supporting leads.

$$F_B = iLB \sin 90^\circ$$

$$F_B = iLB$$

$$F_B = mg$$

$$iLB = mg$$

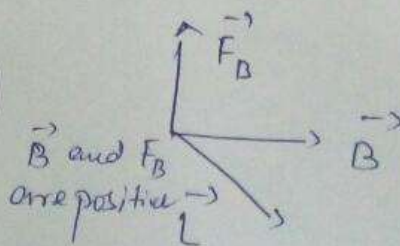
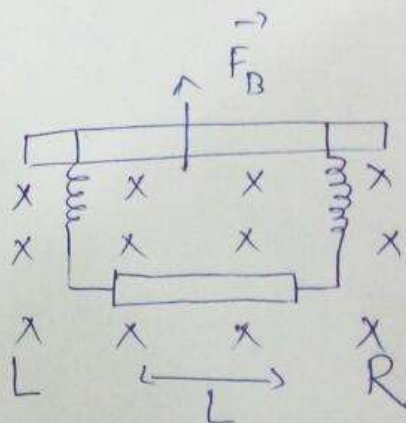
$$i = \frac{mg}{LB}$$

$$(a) |i| = \frac{13 \times 10^{-3} \times 9.8}{62 \times 10^{-2} \times 0.440} = 0.467\text{ A}$$

(b) from R.H.R

current is on x -axis ($+\hat{i}$)

left to right along the direction of $\vec{L}$ length vector



P-40: A wire 1.80m long carries a current of 13A and makes an angle of 35° with a uniform Magnetic field of Magnitude $B=1.5T$. Calculate The Magnetic force on the wire?

Solution :-

$$L = 1.8m$$

$$i = 13A$$

$$\theta = 35^\circ$$

$$B = 1.50T, F_B = ?$$

$$F_B = iLB \sin \theta$$

$$= 13 \times 1.8 \times 1.5 \times \sin 35^\circ$$

$$\boxed{F_B = 20.1 N}$$

P-25. (a) Find the frequency of revolution of an electron with an energy of 100 eV in a uniform magnetic field of Magnitude 35 μ T (b) Calculate the radius of the path of this electron if its velocity is perpendicular to the Magnetic field?

(a) $f = ?$

Energy = 100 eV

$B = 35 \mu$ T

(b) $r = ?$

$\vec{v} \perp \vec{B}$

$S = vt$
as

$T = \frac{2\pi r}{v}$

$T = \frac{S}{v}$

So, $f = \frac{\omega}{2\pi} = \frac{1}{T}$

So, $f = \frac{qB}{2\pi m_e}$

$T = \frac{2\pi r}{v}$

$T = \frac{2\pi}{v} \left(\frac{mv}{qB} \right)$

$\therefore r = \frac{mv}{qB}$

$f = \frac{1.6 \times 10^{-19} \times 35 \times 10^{-6}}{2\pi \times 9.11 \times 10^{-31}} = 9.78 \times 10^5 \text{ Hz}$

$\approx f = 10 \times 10^5 \text{ Hz}$

(b) $r = \frac{mv}{qB}$ — (1)

(1) become

$r = \frac{m}{qB} \sqrt{\frac{2K \cdot E}{m}}$

$r = \frac{\sqrt{2mK \cdot E}}{qB} = \frac{\sqrt{2 \times 9.11 \times 10^{-31} \times 100 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 35 \times 10^{-6}}$

$r = 0.964 \text{ m}$

$r = \frac{mv}{qB}$ (radius).

$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{qB} = \frac{2\pi m}{qB}$ (period).

$f = \frac{1}{T} = \frac{qB}{2\pi m}$ (frequency).

$\omega = 2\pi f = \frac{qB}{m}$ (angular frequency).

$\frac{1}{2} mv^2 = K \cdot E$

$v^2 = \frac{2K \cdot E}{m}$

$v = \sqrt{\frac{2K \cdot E}{m}}$

P-24: An electron is accelerated from rest by a potential difference of 350 V. It then enters a uniform Magnetic field of magnitude 200 mT with its velocity perpendicular to the field. Calculate (a) speed of the electron and (b) the radius of its path in the Magnetic field.

$$P.D = 350 V$$

$$B = 200 \text{ mT} \quad \vec{v} \perp \vec{B}$$

(a) speed of $e^- = ?$

(b) $r =$ radius of its path in the Mag. field

during movement of electron P.E = qV will be equal to K.E = $\frac{1}{2} m_e v^2$

They will balance each other or will be in equilibrium.

$$qV = \frac{1}{2} m_e v^2$$

$$eV = \frac{1}{2} m_e v^2$$

$$1.6 \times 10^{-19} \times 350 = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$$

$$\boxed{v = 1.11 \times 10^7 \text{ m/s}}$$

$$(b) \quad r = \frac{mv}{qB}$$

$$r = \frac{9.11 \times 10^{-31} \times 1.11 \times 10^7}{1.6 \times 10^{-19} \times 200 \times 10^{-3}}$$

$$\boxed{r = 3.1 \times 10^{-4} \text{ m}}$$

P-23: What uniform Magnetic field, applied perpendicular to a beam of electrons having moving at $1.30 \times 10^6 \text{ m/s}$ is Required to make the electrons travel in a circular arc of radius 0.350 m ?

mass of $e^- = 9.1 \times 10^{-31} \text{ kg}$

velocity of $e^- = 1.3 \times 10^6 \text{ m/s}$

radius of circular arc = 0.350 m

$$B = \frac{m_e v}{e r}$$

$$r = \frac{mv}{|q|B} \quad (\text{radius}).$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \frac{mv}{|q|B} = \frac{2\pi m}{|q|B} \quad (\text{period}).$$

$$f = \frac{1}{T} = \frac{|q|B}{2\pi m} \quad (\text{frequency}).$$

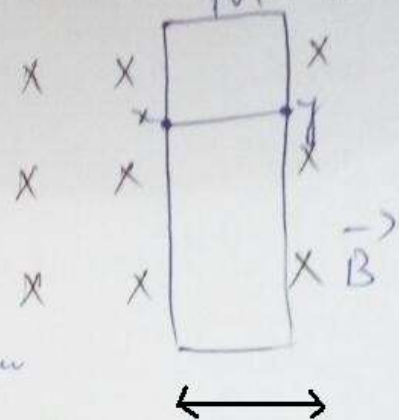
$$\omega = 2\pi f = \frac{|q|B}{m} \quad (\text{angular frequency}).$$

$$\frac{9.1 \times 10^{-31} \times 1.3 \times 10^6}{1.6 \times 10^{-19} \times 0.350}$$

$$B = 2.11 \times 10^{-5} \text{ T}$$

P-14: A Metal strip 6.50m long, 0.850cm wide, and 0.760mm thick moves with constant velocity $\vec{v}$ through a uniform magnetic field $B = 1.20\text{mT}$ directed perpendicular to the strip, as shown in Fig 28.34

A potential difference of $3.90\mu\text{V}$ is measured b/w points x and y across the strip. Calculate the speed V .



potential across the width of metal strip

length of Metal strip : 6.50m

width = 0.850cm = d

thickness = 0.760mm

$B = 1.2\text{mT}$

$\Delta V = 3.90\mu\text{V}$ (b/w x and y)

$v = ?$

$$F_e = F_B$$

$$qE = qvB$$

$$v = E/B$$

$$\text{So, } v = \frac{E}{B}$$

$$\text{and } E = \frac{V}{d}$$

So,

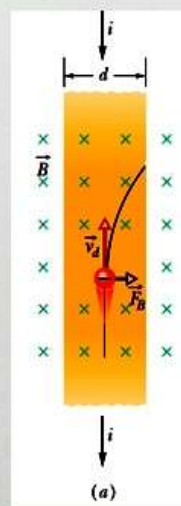
$$v = \frac{E}{B}$$

$$v = \frac{4.58 \times 10^{-4}}{1.2 \times 10^{-3}}$$

$$E = \frac{3.9 \times 10^{-6}}{0.850 \times 10^{-2}}$$

$$E = 4.58 \times 10^{-4} \text{ N/m}$$

$$\boxed{\vec{v} = 0.382 \text{ m/s}}$$



P-13: A strip of copper $150\mu\text{m}$ thick and 4.5mm wide is placed in a uniform magnetic field $\vec{B}$ of magnitude 0.65T , with $\vec{B}$ perpendicular to the strip. A current $i = 23\text{A}$ is then sent through the strip such that a Hall potential difference V appears across the width of the strip. Calculate V . (The Number of charge carriers per unit volume for copper is 8.47×10^{28} electrons/ m^3).

$$\text{Thickness} = l = 150\mu\text{m}$$

$$\text{width} = d = 4.5\text{mm}$$

$$B = 0.65\text{T}$$

$B \perp \text{strip}$

$$i = 23\text{A}$$

Hall potential develops $V = ?$

$$n = 8.47 \times 10^{28} \text{ electrons}/\text{m}^3$$

$$n = \frac{Bi}{Vle}$$

$$8.47 \times 10^{28} = \frac{0.65 \times 23}{V \times (50 \times 10^{-6} \times 1.6 \times 10^{-19})}$$

$$V \times (50 \times 10^{-6} \times 1.6 \times 10^{-19})$$

$$V = \frac{14.75}{2032800} = 7.35 \times 10^{-6}\text{V}$$

$$\boxed{V = 7.35\mu\text{V}}$$

P-5: An electron moves through a uniform magnetic field given by $\vec{B} = B_x \hat{i} + (3.0 B_x) \hat{j}$. At a particular instant, the electron has velocity $\vec{v} = (2.0 \hat{i} + 4.0 \hat{j}) \text{ m/s}$ and the magnetic force acting on it is $(6.4 \times 10^{-19} \text{ N}) \hat{k}$.

Find B_x .

$$\vec{B} = [B_x \hat{i} + (3.0 B_x) \hat{j}] \text{ Tesla} \Rightarrow B_y = 3B_x$$

$$\vec{v} = (2\hat{i} + 4\hat{j}) \text{ m/s}$$

$$\vec{F}_B = (6.4 \times 10^{-19} \text{ N}) \hat{k}, \quad B_x = ?$$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\begin{aligned} \vec{F} &= q \{ (v_x \hat{i} + v_y \hat{j}) \times (B_x \hat{i} + B_y \hat{j}) \} \\ &= q \{ v_x B_x \hat{i} \times \hat{i} + v_x B_y \hat{i} \times \hat{j} + v_y B_x \hat{j} \times \hat{i} + v_y B_y \hat{j} \times \hat{j} \} \end{aligned}$$

$$\vec{F}_B = q (v_x B_y \hat{k} + v_y B_x (-\hat{k}))$$

$$= q (v_x B_y - v_y B_x) \hat{k}$$

$$F_z = F_B = q [v_x (3B_x) - v_y (B_x)] \hat{k}$$

$$F_z = q(B_x)(3v_x - v_y)$$

$$B_x = F_z / [q(3v_x - v_y)] = \frac{6.4 \times 10^{-19}}{-1.6 \times 10^{-19} (3 \times 2 - 4)}$$

$$\boxed{B_x = -2 \text{ Tesla}}$$

P-4: An alpha particle travels at a velocity $\vec{v}$ of magnitude 550 m/s through a uniform magnetic field $\vec{B}$ of magnitude 0.045 T. (An alpha particle has a charge of $+3.2 \times 10^{-19}$ C and a mass of 6.6×10^{-27} kg). The angle b/w $\vec{v}$ and $\vec{B}$ is 52° . What is the magnitude of (a) the force $\vec{F}_B$ acting on the particle due to the field and (b) acceleration of the particle due to $\vec{F}_B$? (c) Does the speed of the particle increase, decrease or remain the same.

α -particle travels at a velocity $\vec{v} = 550 \text{ m/s}$

$\vec{B} = 0.045 \text{ T (uniform)}$

$+q$ of α particle = $+3.2 \times 10^{-19} \text{ C}$

mass " " = $6.6 \times 10^{-27} \text{ kg}$

angle b/w $\vec{v}$ and $\vec{B} = 52^\circ$

$$(a) \quad F_B = q v B \sin \phi = 3.2 \times 10^{-19} \times 550 \times 0.045 \sin 52^\circ$$

$$F_B = 6.2 \times 10^{-18} \text{ N}$$

$$(b) \quad a = \frac{F_B}{m_\alpha} = \frac{6.2 \times 10^{-18}}{6.6 \times 10^{-27}} = 9.5 \times 10^8 \text{ m/s}^2$$

(c) remains the same

constant force (F_B)

So, speed remains the same.

P-3: An electron that has an instantaneous velocity of $\vec{v} = (2.0 \times 10^6 \text{ m/s})\hat{i} + (3.0 \times 10^6 \text{ m/s})\hat{j}$ is moving through the uniform Magnetic field $\vec{B} = (0.030)\hat{i} - (0.15)\hat{j}$. (a) Find the force on the e^- due to the Magnetic field. (b) Repeat your calculation for a proton having the same velocity.

$$\vec{v} = [(2 \times 10^6)\hat{i} + (3 \times 10^6)\hat{j}] \text{ m/s}$$

$$\vec{B} = [(0.030)\hat{i} - (0.15)\hat{j}] \text{ Tesla}$$

(a) $F_B = ?$ on electron due to B

(b) $F_B = ?$ (on proton)

$$\hat{i} \times \hat{i} = 0 \quad \hat{j} \times \hat{j} = 0 \quad \hat{k} \times \hat{k} = 0$$

$$\hat{i} \times \hat{j} = \hat{k} \quad \hat{j} \times \hat{k} = \hat{i} \quad \hat{k} \times \hat{i} = \hat{j}$$

$$\hat{j} \times \hat{i} = -\hat{k} \quad \hat{k} \times \hat{j} = -\hat{i} \quad \hat{i} \times \hat{k} = -\hat{j}$$

$$\vec{F}_B = q\vec{v} \times \vec{B}$$

$$= q \left([(2 \times 10^6)\hat{i} + (3 \times 10^6)\hat{j}] \times [(0.030)\hat{i} - (0.15)\hat{j}] \right)$$

$$\vec{F}_B = -1.6 \times 10^{-19} \left[(2 \times 10^6) \hat{i} \times \hat{i} - (2 \times 10^6)(0.15) \hat{i} \times \hat{j} + (3 \times 10^6)(0.03) \hat{j} \times \hat{i} - (3 \times 10^6)(0.15) \hat{j} \times \hat{j} \right]$$

$$= (-1.6 \times 10^{-19}) [0 - 300,000 \hat{k} + 90,000 (-\hat{k}) - 0]$$

$$= (-1.6 \times 10^{-19}) (-300,000 \hat{k} - 90,000 \hat{k})$$

$$\vec{F}_B = 6.24 \times 10^{-14} \hat{k} \quad (\text{z-direction})$$

(b) $\vec{F}_B = -6.24 \times 10^{-14} \hat{k} \quad (-\text{ve z-direction})$

P-1. A proton travelling at 23.0° with respect to the direction of a Magnetic field strength 2.60 mT experiences a Magnetic force of $6.50 \times 10^{-17} \text{ N}$. Calculate (a) the proton's speed and (b) its Kinetic energy in electron-volts.

$$\phi = 23^\circ ; B = 2.60 \text{ mT}$$

$$F_B = 6.50 \times 10^{-17} \text{ N}$$

Calculate (a) the proton's speed = ?

(b) K.E in eV = ?

$$F_B = qvB \sin \phi = e v B \sin \phi$$

$$v = \frac{F_B}{e B \sin \phi} = \frac{6.5 \times 10^{-17}}{1.6 \times 10^{-19} \times 2.6 \times 10^{-3} \times \sin 23}$$

$$\boxed{v = 4 \times 10^5 \text{ m/s}}$$

$$(b) \text{ K.E} = \frac{1}{2} m_p v^2$$

$$= \frac{1}{2} \times 1.67 \times 10^{-27} \times (4 \times 10^5)^2$$

$$= 1.34 \times 10^{-16} \text{ J}$$

$$\text{K.E} = \frac{1.34 \times 10^{-16}}{1.6 \times 10^{-19}}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$\boxed{\text{K.E} = 835 \text{ eV}}$$

P-45: A wire 50cm long carries a 0.500A current in the positive direction of an x-axis through a magnetic field $\vec{B} = (3.00\text{mT})\hat{j} + (10.0\text{mT})\hat{k}$. In unit vector notation, what is the magnetic force on the wire?

By

Bz

$$L = 50\text{cm}, \quad i = 0.50\text{A} \quad (\text{+ve x-axis})$$

$$\vec{B} = (3\hat{j} + 10\hat{k})\text{mT}$$

$$F_B = ? \quad (\text{In unit vector notation})$$

$$F_B = i \vec{L} \times \vec{B}$$

$$= i L (\hat{i}) \times (B_y \hat{j} + B_z \hat{k})$$

$$= i L B_y \hat{i} \times \hat{j} + i L B_z \hat{i} \times \hat{k}$$

$$= i L [B_y \hat{k} + B_z (-\hat{j})]$$

$$F_B = 0.5 \times 50 \times 10^{-2} [3 \times 10^{-3} \hat{k} - 10 \times 10^{-3} \hat{j}]$$

$$= 0.25 \times 3 \times 10^{-3} \hat{k} - 0.25 \times 10 \times 10^{-3} \hat{j}$$

$$F_B = 7.5 \times 10^{-4} \hat{k} - 2.5 \times 10^{-3} \hat{j}$$

$$\vec{F}_B = (-2.5 \times 10^{-3} \hat{j} + 7.5 \times 10^{-4} \hat{k}) \text{N}$$

